



Autoignition of surrogate biodiesel fuel (B30) at high pressures: Experimental and modeling kinetic study

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ABSTRACT

Ignition delay times of surrogate biodiesel fuels were measured in a high-pressure shock tube over a wide range of experimental conditions (pressures of 20 and 40 bar, equivalence ratios in the range 0.5–1.5, and temperatures ranging from 700 to 1200 K). A detailed chemical kinetic mechanism developed for the oxidation of a biodiesel fuel and a B30 biodiesel surrogate (49% *n*-decane, 21% 1-methylnaphthalene, and 30% methyl octanoate in mol%) was used to simulate the present experiments. Cross reactions between radicals from the three fuel components and reactions of methylnaphthalene oxidation recently proposed in the literature were introduced into the model in order to improve ignition delay time predictions at low temperatures. The new scheme (7865 reversible reactions and 1975 species) yields improved model predictions of concentration profiles measured earlier in a jet-stirred reactor, and also represents fairly well the present experimental data over the entire range of conditions of this study. Sensitivity analyses and reaction path analyses were used to rationalize the results.

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1. Introduction

The optimization of diesel engines relies not only on downsizing the combustion chamber, but also on improving diesel fuel combustion and understanding the complex processes that underlie combustion behavior. Blends of biofuels and fossil fuels provide one option for decreasing both carbon oxide and PAH (polycyclic aromatic hydrocarbon) emissions and dependence on fossil fuels in many countries around the world. In Europe, for example, according to Directive 2003/30/EC, EU member states must ensure that a minimum quantity of biofuel and other renewable fuels is placed on their markets. For this purpose, a reference value of 10%, calculated on energy content, of all hydrocarbon transportation fuels must be replaced by the end of 2020 [1]. However, blends of 30% biodiesel with 70% conventional diesel (B30) have different properties than conventional diesel fuels and generally require engine modifications.

As a conventional diesel fuel, biodiesel (or agro-diesel) consists of a complex multicomponent mixture of fatty acid methyl esters (FAME) produced from transesterification of triglycerides (oil) with methanol. Current biodiesel fuels are mixtures of ca. C₁₂–C₂₂ highly saturated carbon-chain esters. Therefore, to sim-

plify the description of the chemical properties of these complex mixtures, surrogates with a limited number of analogous components are used to simulate biodiesel combustion kinetics. The B30 surrogate studied here contains an *n*-paraffinic compound with a long carbon chain, which represents the paraffinic fraction of conventional diesel fuel; a diaromatic hydrocarbon, which plays an important role in soot formation reactions; and a methyl ester with a relatively long carbon chain, which is the common chemical structure observed in the complex composition of FAME. The B30 surrogate fuel (49% *n*-decane, 21% 1-methylnaphthalene and 30% methyl octanoate in mol%) has properties similar to those of a conventional B30 fuel, i.e., a normal density of 0.82 g/l at 20 °C, a cetane number (CN) of ≈48, and a hydrogen-to-carbon ratio of 1.84.

In this study, a validated detailed chemical kinetic mechanism of *n*-decane/1-methylnaphthalene/methyl octanoate, proposed earlier [2], is tested against high- to low-temperature/high-pressure ignition delay time data for a surrogate B30. These experiments were performed in a shock tube equipped with a heating system to (1) evaluate how accurately the validated detailed kinetic mechanism can predict ignition delay times at high pressure over a wide range of temperatures, (2) improve the kinetic modeling of low-temperature oxidation, where cross reactions between large fuel radicals may play an important role, and (3) provide a kinetic model, which is able to predict real combustion behavior within the bounds of a larger experimental dataset.

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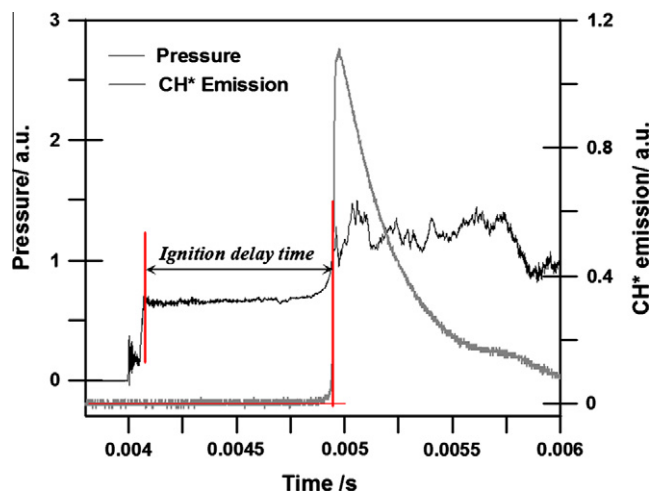


Fig. 1. Pressure history and CH* emission profiles indicating ignition delay for a B30 surrogate-fuel/air mixture at 40 bar, 916 K, and $\phi = 1.0$.

Table 1

Reactions modified in this model ($k = A \cdot T^n \cdot \exp(-E/RT)$ in units of cal, mol, cm^3 , and s).

Reactions	A	n	E	Ref.
$\text{phCH}_2 + \text{O}_2 \rightarrow \text{phHCO} + \text{OH}$	3.760E+15	-1.55	11,322	[12]
$\text{phCH}_2 + \text{O}_2 \rightarrow \text{C}_6\text{H}_5\text{O} + \text{CH}_2\text{O}$	6.260E+37	-8.86	16,582	[12]
$\text{phCH}_3\text{CH}_2 + \text{O} \rightarrow \text{phCH}_3\text{HCO} + \text{H}$	4.370E+18	-1.34	1592	[12]
$\text{phCH}_3\text{CH}_2 + \text{O} \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CH}_2\text{O}$	5.990E+23	-2.47	16,192	[12]
$\text{phCH}_3\text{CH}_2 + \text{O} \rightarrow \text{toluene} + \text{HCO}$	1.970E+22	-2.36	8152	[12]
$\text{phCH}_3\text{CH}_2 + \text{O}_2 \rightarrow \text{phCH}_3\text{HCO} + \text{OH}$	1.380E+02	2.42	7440	[12]
$\text{phCH}_3\text{CH}_2 + \text{O}_2 \rightarrow \text{OC}_6\text{H}_4\text{CH}_3 + \text{CH}_2\text{O}$	6.570E+03	1.87	5002	[12]
$\text{phCH}_3\text{CH}_3 = \text{C}_6\text{H}_4\text{CH}_3 + \text{CH}_3$	4.320E+29	-3.58	110,165	[12]
$\text{phCH}_3\text{CH}_2 + \text{H} = \text{C}_6\text{H}_4\text{CH}_3 + \text{CH}_3$	5.830E+67	-14.15	68,329	[12]
$\text{phCH}_3\text{CH}_2 + \text{HO}_2 \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CH}_2\text{O} + \text{OH}$	3.130E+18	-1.44	13,944	[12]
$\text{phCH}_3\text{HCO} \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{H}$	2.100E+16	0.00	81,740	[12]
$\text{phCH}_3\text{HCO} + \text{H} \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{H}_2$	4.090E+09	1.16	2404	[12]
$\text{phCH}_3\text{HCO} + \text{O} \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{OH}$	5.840E+12	0.00	1809	[12]
$\text{phCH}_3\text{HCO} + \text{OH} \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{H}_2\text{O}$	2.890E+08	1.35	-1573	[12]
$\text{phCH}_3\text{HCO} + \text{O}_2 \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{HO}_2$	1.200E+05	2.50	37,555	[12]
$\text{phCH}_3\text{HCO} + \text{HO}_2 \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{H}_2\text{O}_2$	4.090E+04	2.50	10,203	[12]
$\text{phCH}_3\text{HCO} + \text{CH}_3 \rightarrow \text{C}_6\text{H}_4\text{CH}_3 + \text{CO} + \text{CH}_4$	3.490E-08	6.21	1630	[12]
$\text{C}_{10}\text{H}_7\text{CH}_2 + \text{O}_2 \rightarrow \text{C}_{10}\text{H}_7\text{HCO} + \text{OH}$	3.760E+15	-1.55	11,322	[12]
$\text{C}_{10}\text{H}_7\text{CH}_2 + \text{O}_2 \rightarrow \text{A2-X} + \text{CH}_2\text{O} + \text{O}$	6.320E+11	0.00	14,500	Estd. from [12]
$\text{C}_{10}\text{H}_7\text{CH}_2 + \text{HO}_2 \rightarrow \text{C}_{10}\text{H}_7\text{CH}_2\text{O} + \text{OH}$	2.000E+11	0.00	00	Estd. from [12]
$\text{C}_{10}\text{H}_7\text{CH}_2\text{O} \rightarrow \text{A2-X} + \text{CH}_2\text{O}$	1.300E+14	0.00	1100	Estd.
$\text{Indenoxo} + \text{H} \rightarrow \text{indenyl} + \text{OH}$	1.500E+13	0.00	00	Estd.
$\text{phCH}_2\text{O} + \text{OH} \rightarrow \text{phHCO} + \text{H}_2\text{O}$	1.716E+09	1.18	-447	Estd. from [19]
$\text{Acrol} \rightarrow \text{C}_2\text{H}_4 + \text{CO}$	2.450E+12	0.00	67,128	Estd.
$\text{A}_2\text{OH} + \text{O} \rightarrow \text{indenoxo} + \text{HCO}$	4.210E+13	0.00	2410	Estd.
$\text{HCCO} + \text{O}_2 \rightarrow \text{CO}_2 + \text{CO} + \text{H}$	4.780E+12	-0.42	1150	[15]
$\text{HCCO} + \text{OH} \rightarrow \text{C}_2\text{O} + \text{H}_2\text{O}$	3.000E+13	0.00	00	[16]
$\text{OC}_6\text{H}_4\text{CH}_3 \rightarrow \text{C}_6\text{H}_6 + \text{H} + \text{CO}$	2.510E+11	0.00	43,900	Estd. from [12]
$\text{C}_{10}\text{H}_{20}\text{qjBqA} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjCqA} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjDqA} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjEqA} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjAqB} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjCqB} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjDqB} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjEqB} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjFqB} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjAqC} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjBqC} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjDqC} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjEqC} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjFqC} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjGqC} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjAqD} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjBqD} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjCqD} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]
$\text{C}_{10}\text{H}_{20}\text{qjEqD} \rightarrow \text{CH}_3\text{O} + 3\text{C}_2\text{H}_4 + \text{C}_2\text{H}_5 + \text{OH} + \text{CO}_2$	7.000E+15	0.00	42,065	[18]

(continued on next page)

2. Experimental

The experiments were carried out in a high-pressure shock tube, heated to 403 K, similar to conditions employed in previous works by Schulz and co-workers [3]. The shock tube has an internal diameter of 90 mm, with a driven section of 6.1 m and a driver section of 6.4 m in length. The driven and driver sections are separated by an aluminum diaphragm. The facility is equipped with a pumping system to reach pressures below 10^{-2} mbar between experiments and three piezoelectric pressure gauges to measure shock wave speed. Fuel/air gas mixtures were prepared at 403 K in a stainless steel mixing vessel before being filled into the driven section. The mixtures were prepared in the mixing vessel for each set of experiments not exceeding 5 h of utilization (residence time) after preparation. Every day fresh mixture was prepared and used. We assume that the residence time of 5 h at 403 K is not time enough to produce any preoxidation or pyrolysis prior to the experiment. Pressures of evaporated fuel and synthetic air (79.5% N_2 and 20.5% O_2) were controlled manometrically to ensure the desired

Table 1 (continued)

Reactions	A	n	E	Ref.
$C_{10}H_{20}qjFqD \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjGqD \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjHqD \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjAqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjBqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjCqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjDqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjFqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjGqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjHqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]
$C_{10}H_{20}qjIqE \rightarrow CH_3O + 3C_2H_4 + C_2H_5 + OH + CO_2$	7.000E+15	0.00	42,065	[18]

Note: “ph” stands for C_6H_5 . The nomenclature was already introduced in previous work [7,9] and is presented in Supplementary information.

Table 2

Cross reactions considered in this model ($k = A \cdot T^n \cdot \exp(-E/RT)$ in units of cal, mol, cm^3 , and s).

Reactions	A	n	E	Ref.
$C_6H_5OH + C_2H_2 \rightarrow A\text{-styryl} + OH$	3.210E+13	0.00	8910	Estd.
$C_6H_6 + mo8j \rightarrow mo + C_6H_5$	3.000E+12	0.00	14,000	Estd. from [20]
$C_6H_6 + mo7j \rightarrow mo + C_6H_5$	3.000E+12	0.00	14,000	Estd. from [20]
$C_6H_6 + mo6j \rightarrow mo + C_6H_5$	3.000E+12	0.00	14,000	Estd. from [20]
$C_6H_6 + mo5j \rightarrow mo + C_6H_5$	3.000E+12	0.00	14,000	Estd. from [20]
$C_6H_6 + mo4j \rightarrow mo + C_6H_5$	3.000E+12	0.00	14,000	Estd. from [20]
$A_2OH + C_2H_6 \rightarrow naphthalene + OH + C_2H_5$	18.210E+14	0.00	16,410	Estd. from [20]
$AC_7H_{15} + toluene \rightleftharpoons phCH_2 + nC_7H_{16}$	1.000E+11	0.00	12,000	[20]
$BC_7H_{15} + toluene \rightleftharpoons phCH_2 + nC_7H_{16}$	1.000E+11	0.00	12,000	[20]
$CC_7H_{15} + toluene \rightleftharpoons phCH_2 + nC_7H_{16}$	1.000E+11	0.00	12,000	[20]
$DC_7H_{15} + toluene \rightleftharpoons phCH_2 + nC_7H_{16}$	1.000E+11	0.00	12,000	[20]
$C_7H_{15}qjA + toluene \rightleftharpoons C_7H_{15}qA + phCH_2$	4.000E+11	0.00	14,000	[20]
$C_7H_{15}qjB + toluene \rightleftharpoons C_7H_{15}qC + phCH_2$	4.000E+11	0.00	14,000	[20]
$C_7H_{15}qjD + toluene \rightleftharpoons C_7H_{15}qD + phCH_2$	4.000E+11	0.00	14,000	[20]
$C_6H_5 + nC_7H_{16} \rightleftharpoons AC_7H_{15} + C_6H_6$	9.000E+11	0.00	15,000	Estd. from [20]
$C_6H_5 + nC_7H_{16} \rightleftharpoons BC_7H_{15} + C_6H_6$	9.000E+11	0.00	15,000	Estd. from [20]
$C_6H_5 + nC_7H_{16} \rightleftharpoons CC_7H_{15} + C_6H_6$	9.000E+11	0.00	15,000	Estd. from [20]
$C_6H_5 + nC_7H_{16} \rightleftharpoons DC_7H_{15} + C_6H_6$	9.000E+11	0.00	15,000	Estd. from [20]
$C_7H_{14}\text{-}1 + phCH_2 \rightleftharpoons CC_7H_{13}\text{-}1 + toluene$	1.000E+11	0.00	17,000	[20]
$C_7H_{14}\text{-}2 + phCH_2 \rightleftharpoons DC_7H_{13}\text{-}1 + toluene$	1.000E+11	0.00	17,000	[20]
$C_7H_{14}\text{-}3 + phCH_2 \rightleftharpoons AC_7H_{13}\text{-}3 + toluene$	1.000E+11	0.00	17,000	[20]
$C_{10}H_{20}\text{-}1 + C_{10}H_7CH_2 \rightleftharpoons CC_{10}H_{19}\text{-}1 + C_{10}H_7CH_3$	1.000E+11	0.00	17,000	Estd. from [20]
$C_{10}H_{20}\text{-}2 + C_{10}H_7CH_2 \rightleftharpoons DC_{10}H_{19}\text{-}2 + C_{10}H_7CH_3$	1.000E+11	0.00	17,000	Estd. from [20]
$C_{10}H_{20}\text{-}3 + C_{10}H_7CH_2 \rightleftharpoons EC_{10}H_{19}\text{-}3 + C_{10}H_7CH_3$	1.000E+11	0.00	17,000	Estd. from [20]
$C_{10}H_{20}\text{-}4 + C_{10}H_7CH_2 \rightleftharpoons FC_{10}H_{19}\text{-}4 + C_{10}H_7CH_3$	1.000E+11	0.00	17,000	Estd. from [20]
$C_7H_{15}qjA + phHCO \rightleftharpoons phCO + C_7H_{15}qA$	3.000E+12	0.00	9000	[20]
$C_7H_{15}qjB + phHCO \rightleftharpoons phCO + C_7H_{15}qA$	3.000E+12	0.00	9000	[20]
$C_7H_{15}qjC + phHCO \rightleftharpoons phCO + C_7H_{15}qA$	3.000E+12	0.00	9000	[20]
$C_7H_{15}qjD + phHCO \rightleftharpoons phCO + C_7H_{15}qA$	3.000E+12	0.00	9000	[20]
$AC_8H_{17} + toluene \rightleftharpoons phCH_2 + nC_8H_{18}$	2.000E+11	0.00	14,000	Estd. from [20]
$BC_8H_{17} + toluene \rightleftharpoons phCH_2 + nC_8H_{18}$	1.000E+11	0.00	12,000	Estd. from [20]
$CC_8H_{17} + toluene \rightleftharpoons phCH_2 + nC_8H_{18}$	1.000E+11	0.00	12,000	Estd. from [20]
$DC_8H_{17} + toluene \rightleftharpoons phCH_2 + nC_8H_{18}$	1.000E+11	0.00	12,000	Estd. from [20]
$C_6H_5 + nC_8H_{18} \rightleftharpoons AC_8H_{17} + C_6H_6$	9.000E+11	0.00	15,500	Estd. from [20]
$C_6H_5 + nC_8H_{18} \rightleftharpoons BC_8H_{17} + C_6H_6$	9.000E+11	0.00	15,500	Estd. from [20]
$C_6H_5 + nC_8H_{18} \rightleftharpoons CC_8H_{17} + C_6H_6$	9.000E+11	0.00	15,500	Estd. from [20]
$C_6H_5 + nC_8H_{18} \rightleftharpoons DC_8H_{17} + C_6H_6$	9.000E+11	0.00	15,500	Estd. from [20]
$C_{10}H_{21}ojA + A_2OH \rightarrow naphthalene + C_{10}H_{20}rDqA$	3.000E+12	0.00	00	Estd. from [20]
$C_{10}H_{21}ojB + A_2OH \rightarrow naphthalene + C_{10}H_{20}rAqB$	3.000E+12	0.00	00	Estd. from [20]
$C_{10}H_{21}ojC + A_2OH \rightarrow naphthalene + C_{10}H_{20}rAqC$	3.000E+12	0.00	00	Estd. from [20]
$C_{10}H_{21}ojD + A_2OH \rightarrow naphthalene + C_{10}H_{20}rAqD$	3.000E+12	0.00	00	Estd. from [20]
$C_{10}H_{21}ojE + A_2OH \rightarrow naphthalene + C_{10}H_{20}rAqE$	3.000E+12	0.00	00	Estd. from [20]
$A_2OH + AC_{10}H_{21} \rightarrow naphthalene + OH + C_{10}H_{20}\text{-}1$	1.000E+11	0.00	9500	[20]
$A_2OH + BC_{10}H_{21} \rightarrow naphthalene + OH + C_{10}H_{20}\text{-}2$	1.000E+11	0.00	9500	[20]
$A_2OH + CC_{10}H_{21} \rightarrow naphthalene + OH + C_{10}H_{20}\text{-}3$	1.000E+11	0.00	9500	[20]
$A_2OH + DC_{10}H_{21} \rightarrow naphthalene + OH + C_{10}H_{20}\text{-}4$	1.000E+11	0.00	9500	[20]
$A_2OH + DC_{10}H_{21} \rightarrow naphthalene + OH + C_{10}H_{20}\text{-}5$	1.000E+11	0.00	9500	[20]
$C_{10}H_{21}qjA + phHCO \rightarrow phCO + C_{10}H_{21}qA$	3.000E+12	0.00	9000	Estd. from [20]
$C_{10}H_{21}qjB + phHCO \rightarrow phCO + C_{10}H_{21}qB$	3.000E+12	0.00	9000	Estd. from [20]
$C_{10}H_{21}qjC + phHCO \rightarrow phCO + C_{10}H_{21}qC$	3.000E+12	0.00	9000	Estd. from [20]
$C_{10}H_{21}qjD + phHCO \rightarrow phCO + C_{10}H_{21}qD$	3.000E+12	0.00	9000	Estd. from [20]
$C_{10}H_{21}qjE + phHCO \rightarrow phCO + C_{10}H_{21}qE$	3.000E+12	0.00	9000	Estd. from [20]
$A_2R_5 + O_2 \rightarrow A_2\text{-}X + 2CO + H$	9.570E+20	-2.27	7189	Estd. from [12]
$Napht + O_2 \rightarrow indenox + CO + H$	3.000E+13	0.00	8982	Estd. from [12]
$memj + O_2 \rightleftharpoons memOO$	4.520E+12	0.00	00	[17]
$C_9H_{19}CO + O_2 \rightleftharpoons C_9H_{19}COOO$	4.520E+12	0.00	00	[17]

Note: “ph” stands for C_6H_5 . The nomenclature was already introduced in previous work [7,9] and is presented in Supplementary information.

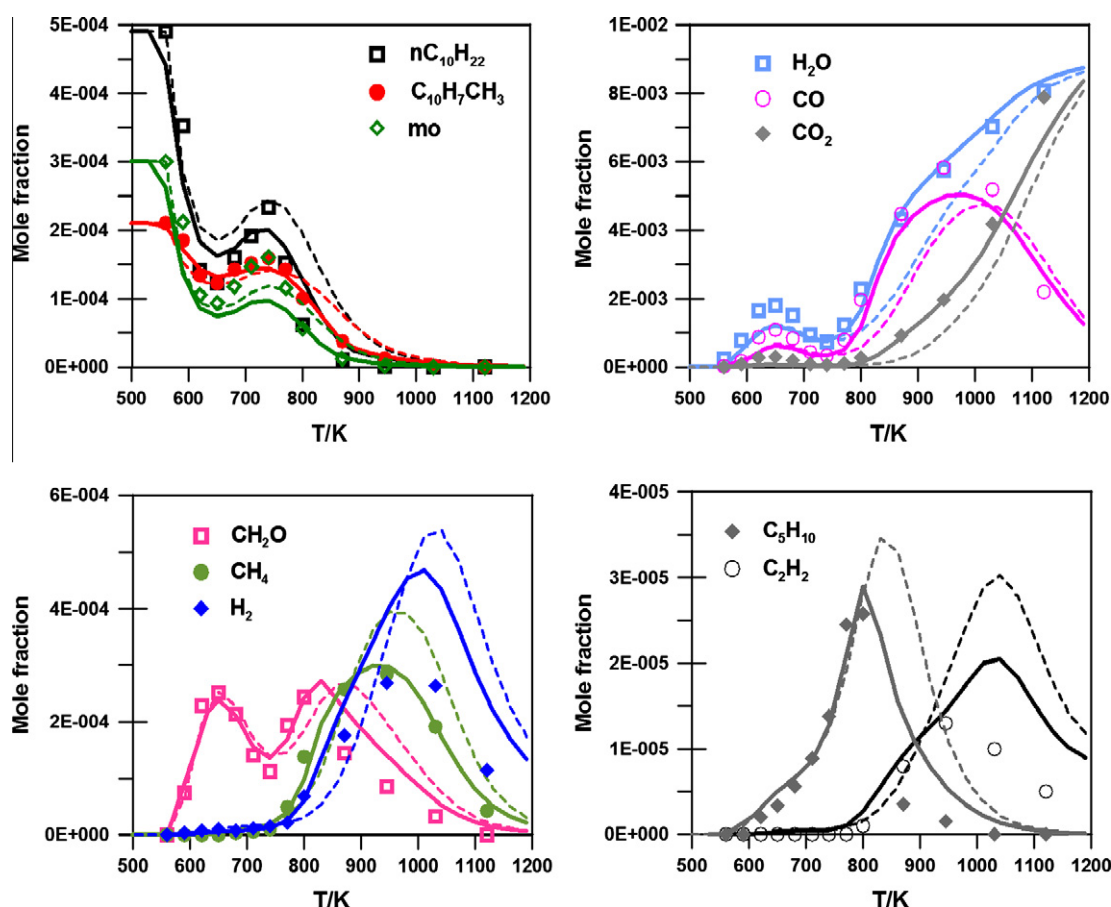


Fig. 2. The B30 surrogate fuel oxidation in a JSR at 10 bar, $\tau = 1$ s, and $\phi = 1$. Symbols: experimental results from a previous study [2]. The estimated uncertainties are in the range of $\pm 10\%$. Solid lines: simulations based on the mechanism with added reactions (Tables 1 and 2). Dashed lines: simulation based on previous mechanism [2].

Table 3

Measured ignition delay times of the B30 surrogate fuel in air at $\phi = 0.5$, $\phi = 1.0$, and $\phi = 1.5$.

$\phi = 0.5$			$\phi = 1.0$			$\phi = 1.5$		
T (K)	p (bar)	τ (μ s)	T (K)	p (bar)	τ (μ s)	T (K)	p (bar)	τ (μ s)
1483	19.6	n.a.	1202	20.0	126	1284	19.5	37
1198	19.9	231	1154	18.2	197	1186	19.4	160
1078	19.0	812	1077	19.0	493	1086	19.3	349
1048	19.9	802	1053	20.2	497	1013	18.2	797
939	19.5	2856	1027	18.9	738	1004	18.0	735
920	18.4	3119	964	15.9	1695	933	19.0	1392
889	20.6	2486	938	19.3	1769	919	20.1	1741
884	19.4	2564	840	19.4	1356	857	18.6	856
865	21.0	2536	827	18.5	1429	835	19.0	883
790	21.6	2205	809	17.4	1626	789	21.5	1189
769	21.4	2592	768	21.4	1312	750	20.0	1631
1215	41.3	112	1456	54.0	n.a.	1887	133.0	n.a.
1178	47.1	229	1275	46.2	n.a.	1145	45.0	n.a.
1113	46.2	79	1210	50.0	23	978	33.0	302
1067	37.1	317	1131	43.0	63	975	28.9	482
1019	42.9	392	1034	38.5	210	889	37.4	433
1000	35.4	781	965	41.8	434	788	41.2	347
977	33.3	1081	928	37.6	395			
957	40.8	781	915	40.6	435			
940	34.6	1328	882	40.2	458			
916	36.3	874	843	35.3	394			
857	37.2	931	814	31.8	792			
840	38.6	876	811	31.5	793			
785	43.3	1114	768	21.4	40			

Note: n.a. indicates ignition occurred after incident shock wave.

equivalence ratio. High-purity hydrocarbons from Aldrich were used: *n*-decane (>99% pure), 1-methylnaphthalene (>95% pure), and methyloctanoate (99% pure). Ar/He driver gas was used to match the acoustic impedance of the test gas, following the methodology of Oertel [4] and Palmer and Knox [5].

The temperature and pressure behind the reflected shock wave were computed from the measured incident shock speed and the attenuation using a one-dimensional shock-tube model (shock-tube code of the CHEMKIN package [6]). The experiments were performed for a post-reflected-shock temperature in the range $700 < T_5 < 1200$ K, with a time resolution of 0.1 μ s, at two different post-reflected shock pressures (20 and 40 bar). The ignition was observed by measuring changes in CH^+ chemiluminescence intensity at 431.5 nm with a photomultiplier 15 mm upstream of the end flange. Ignition delay times (τ_{ign}) were determined by extrapolating the steepest increase of the emission signal to its zero level on the time axis (Fig. 1). Shock speeds were determined by using the signals of three piezo pressure transducers placed over the last 1.0 m of the shock tube and extrapolated to the end wall section. Shock attenuations were typically 0.7–2.5%/m. The measured shock velocities were compared to the initially estimated values of incident shock waves and differences of about $\pm 0.3\%$ were found. Afterward, reflected shock conditions were determined from the standard one-dimensional shock relations and the thermodynamic database using the SHOCK package of CHEMKIN [6]. Uncertainties of about $\pm 0.3\%$ in the incident wave speed yield variations in the calculated T_5 of about ± 7 K and variations in the calculated ignition

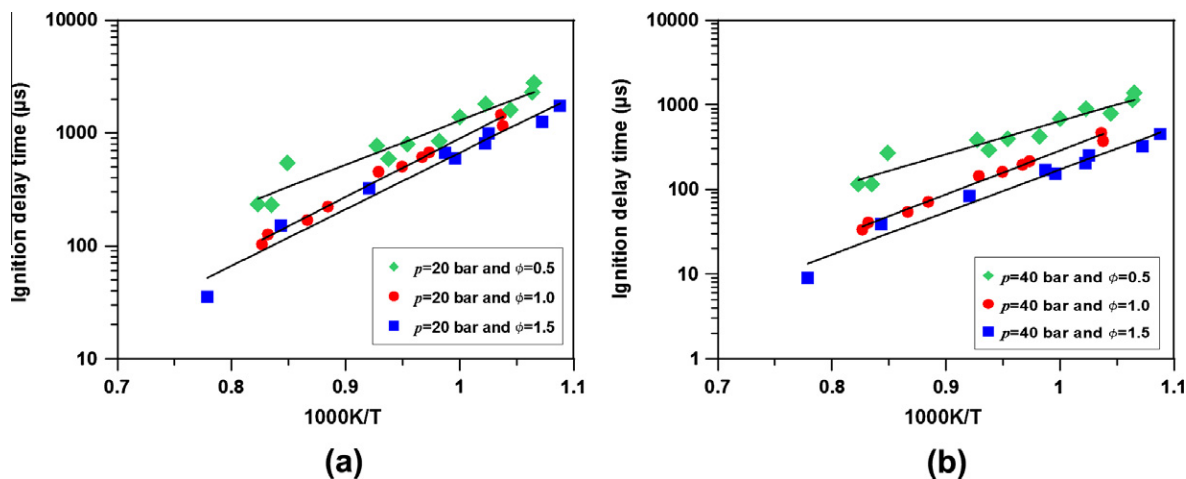


Fig. 3. Normalized ignition delay times for B30 surrogate fuel in air at equivalence ratios of 0.5 (diamonds), 1.0 (circles), and 1.5 (squares), and at pressures ≈ 20 (a) and 40 bar (b). Lines are the Arrhenius fits to the normalized data.

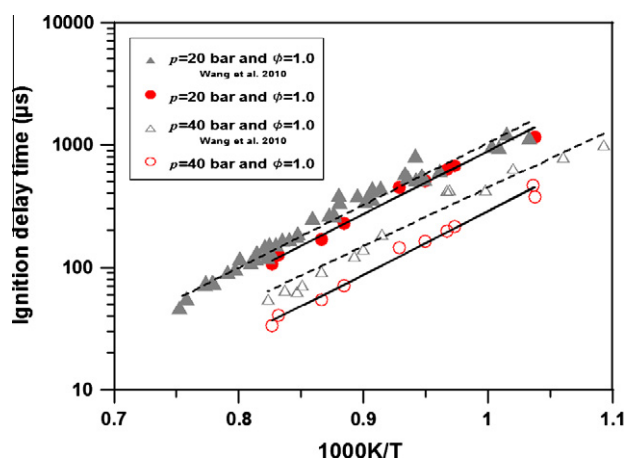


Fig. 4. Effect of temperature and pressure on the ignition of a B30 surrogate fuel containing 30% methyloctanoate, and comparison with the ignition of a diesel surrogate. Triangles: experimental results for 70/30% in mole%, *n*-decane/1-methylnaphthalene mixture in air from [21] (20 bar: solid symbols; 40 bar: open symbols). Circles: experimental results for B30 surrogate fuel in air (20 bar: solid symbols; 40 bar: open symbols). Lines are the Arrhenius fits.

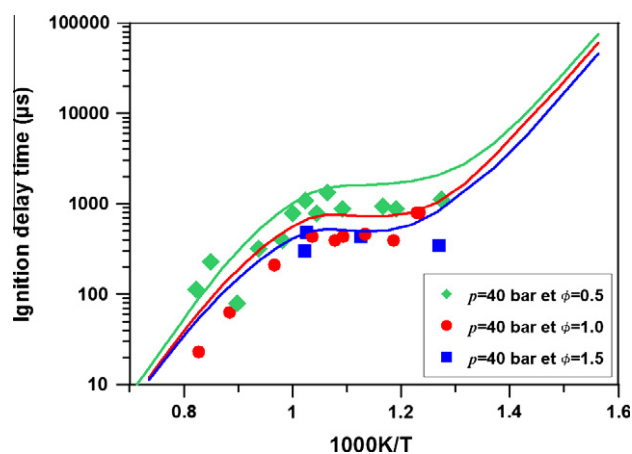


Fig. 6. Ignition delay times for B30 surrogate fuel/air mixtures at 40 bar. Diamonds: experimental data at $\phi = 0.5$. Circles: experimental data at $\phi = 1.0$. Squares: experimental data at $\phi = 1.5$. Lines are the model predictions at different equivalence ratios.

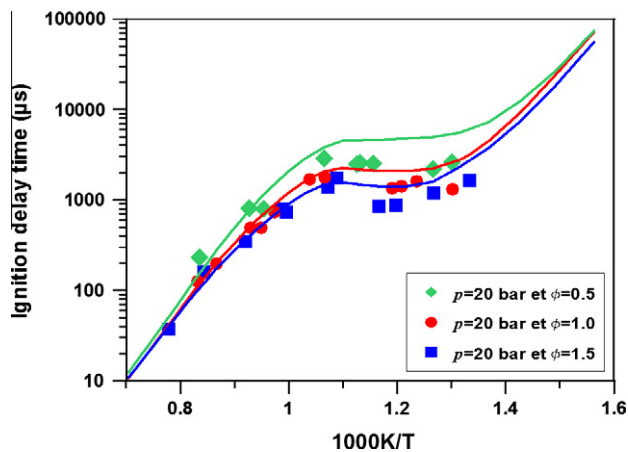


Fig. 5. Ignition delay times for B30 surrogate fuel/air mixtures at 20 bar. Diamonds: experimental data at $\phi = 0.5$. Circles: experimental data at $\phi = 1.0$. Squares: experimental data at $\phi = 1.5$. Lines are the model predictions at different equivalence ratios.

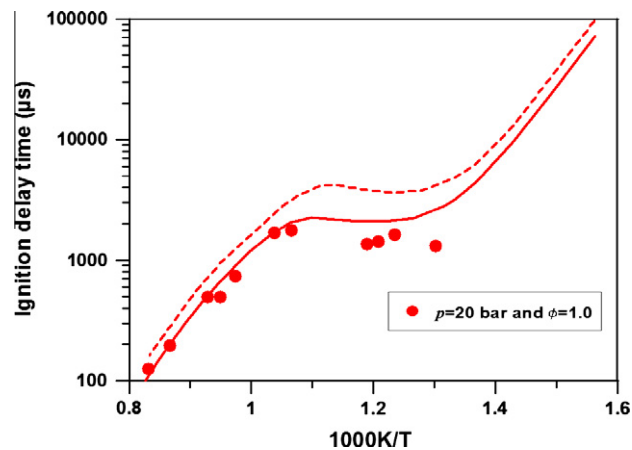


Fig. 7. Ignition delay times for B30 surrogate fuel/air mixtures at $\phi = 1.0$ and 20 bar. Solid lines: Dashed lines: simulation based on previous mechanism [11].

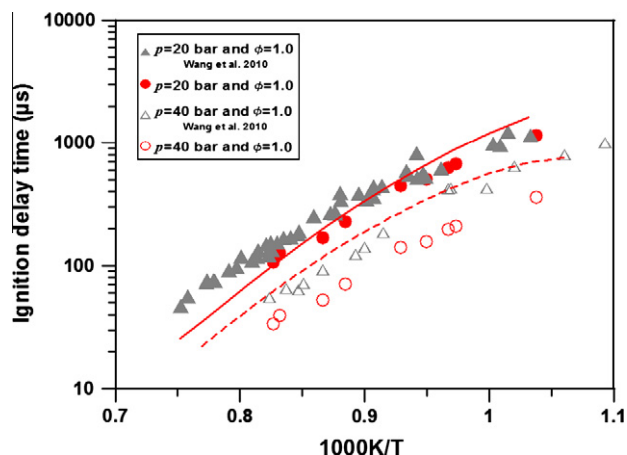


Fig. 8. Comparison between simulated and measured ignition delays of B30 and diesel surrogates. Triangles: experimental results for 70/30% (by mol%) *n*-decane/1-methylnaphthalene mixtures in air from [21] (20 bar full symbols; 40 bar empty symbols). Circles: experimental results for B30 surrogate fuel in air (20 bar full symbols; 40 bar empty symbols). Lines are the model predictions at 20 (solid line) and 40 bar (dashed line).

delay time of about $\pm 7.5\%$, in agreement with the experimental uncertainties (temperature and ignition delay time) found in recent literature for shock tube experiments. In this work, the estimated uncertainties in the ignition delay time are in the range from ± 7.5 to $\pm 10\%$.

3. Modeling

The starting chemical kinetic mechanism was developed in a recent work [2]. It is based on previous studies of the oxidation of methyloctanoate [7], large alkanes [8], and 1-methylnaphthalene [9]. That model, involving 7748 reactions and 1964 species, was validated against JSR (jet stirred reactor) data and showed reasonable agreement with the present experimental results. To that initial version of the mechanism, a small number of cross reactions describing the interactions between fuel components, *n*-decane ($C_{10}H_{22}$), and 1-methylnaphthalene ($C_{10}H_7CH_3$) [10] were added to model the oxidation of commercial and surrogate diesel fuels. These modifications included metathesis reactions of *n*-decane ($C_{10}H_{22}$) with phenyl ($phCH_2$), benzyl (C_6H_5), 1-naphthylmethyl ($C_{10}H_7CH_2$), 1-naphthyl ($A2-X$), and indenyl radicals. Reactions of $C_{10}H_{21}O_2$ ($C_{10}H_{21}qjX$) with toluene, 1-methylnaphthalene ($C_{10}H_7CH_3$), 1-naphthylmethyl ($C_{10}H_7CH_2$), 1-naphthaldehyde ($C_{10}H_7CH_2O$), benzyl (C_6H_5), phenyl ($phCH_2$), and 1-naphthyl ($A2-X$) were also added. Finally, reactions of 1-vinylnaphthalene ($C_{10}H_7C_2H_3$) with HO_2 and reactions of decyl radicals ($C_{10}H_{21}$) with 1-naphthaldehyde ($C_{10}H_7CH_2O$) were added. The kinetics of these added reactions was assumed on the basis of similar reactions in the kinetic base set and from the published work of Andrae et al. [11].

The new chemical kinetics mechanism includes new reactions for aromatics [12] and cross-reactions selected from the literature. The inclusion of these reactions has improved simulations of both concentration profiles in the jet-stirred reactor, and ignition delay times in shock tube. Andrae et al. [11] and Reich and Stivala [13] have demonstrated that the addition of cross reactions must be considered when the surrogate components are very similar, because the chemical interactions between fuel components and their pool of radicals take place during the compression process, from low temperature to ignition temperature, accelerating the ignition processes at low temperature.

The computations were performed using the SENKIN computer code [6]. Reactions of benzyl (C_6H_5), xylene ($phCH_3CH_3$), xylyl

($phCH_3CH_2$), tolualdehyde ($phCH_3HCO$), naphthalene (napht), 1-naphthylmethyl ($C_{10}H_7CH_2$), 1-methylnaphthalene ($C_{10}H_7CH_3$), and acenaphthylene (A_2R_5), based on the Narayanaswamy et al. aromatic mechanism [12], were also considered in this modeling. They included reactions of benzyl with oxygen to produce benzaldehyde ($phHCO$) and phenoxy (C_6H_5O), reactions of xylyl ($phCH_3CH_2$) with oxygen to produce tolualdehyde ($phCH_3HCO$), toluene (toluene), methylphenyl ($phCH_2$), and benzoxy ($phCH_2O$), xylene decomposition, and reactions of benzaldehyde ($phHCO$) with H , O , OH , O_2 , HO_2 , and CH_3 . Also, the kinetics of reactions of $HCCO$ were modified based on the Klippenstein et al. results ($HCCO + O_2 = CO_2 + CO + H$) [15] and the Marinov et al. model ($HCCO + OH = C_2O + H_2O$) [16]. The reactions of memj ($CH_3C(=O)OCH_2$) and $C_9H_{19}CO$ based on the biodiesel fuel surrogate mechanism [17] and the decomposition reactions of $C_{10}H_{20}OOH$ ($C_{10}H_{20}qjBqA$) based on the *n*-decane oxidation mechanism of Bikas and Peters [18] were considered. The kinetics of other reactions were estimated by analogy with similar reactions (Table 1). These reactions have mainly introduced changes in the reaction pathways at $T > 900$ K.

More cross reactions describing interactions between stable intermediate species and the fuel components were also added (Table 2). In the present study, results from shock-tube experiments have shown the strong influence of cross reactions at low temperature ($T < 900$ K) thanks to the good resolution of these experiments over this range of temperatures. These cross reactions accelerate oxidation processes at low temperatures where alkyl radicals have a higher concentration. However, the effect of cross reactions was less important under the conditions of our previous work [2] due to the small changes caused by alkyl radical H-abstractions at high temperature [20].

In the present study, we propose a kinetic mechanism consisting of 7865 reversible reactions involving 1975 species that represents well the kinetics of ignition and oxidation of diesel/biodiesel fuel mixtures. It is available from the authors and as [Supplementary material](#). The rate constants for the reverse reactions were computed from the forward rate constants and the equilibrium constants were calculated using the appropriate thermochemical data [2,10,12,20].

4. Results and discussions

4.1. Results of jet-stirred reactor predictions

In our recent study [2], we measured the kinetics of oxidation of a commercial B30 (30% FAME by vol.) biodiesel fuel and a B30 biodiesel surrogate fuel (49% *n*-decane, 21% 1-methylnaphthalene, 30% methyloctanoate, in mol%) using a pressurized fused-silica jet-stirred reactor under the same initial experimental conditions (560–1030 K, 6 and 10 bar, equivalence ratios of 0.25–1.5). The results of this series of experiments consisted of concentration profiles of reactants, stable intermediates, and products measured as a function of temperature, at fixed residence time (Fig. 2).

Reactions and rate constants were taken or estimated based on analogies with similar reactions in previous works [12,16–18,20]. The first set of reactions (Table 1) involving stable aromatic intermediates were essential to improve the agreement between experimental data and predictions of the current model. In Fig. 2, the concentration profiles simulated with the entire set of reactions are compared with those of the initial model predictions. It can be observed that computed profiles using the new mechanism match the experimental profiles much better than simulations based on our previous kinetic scheme. Similar agreement was observed for the other equivalence ratios. It should be noted that the present model was also successfully tested for the oxidation of

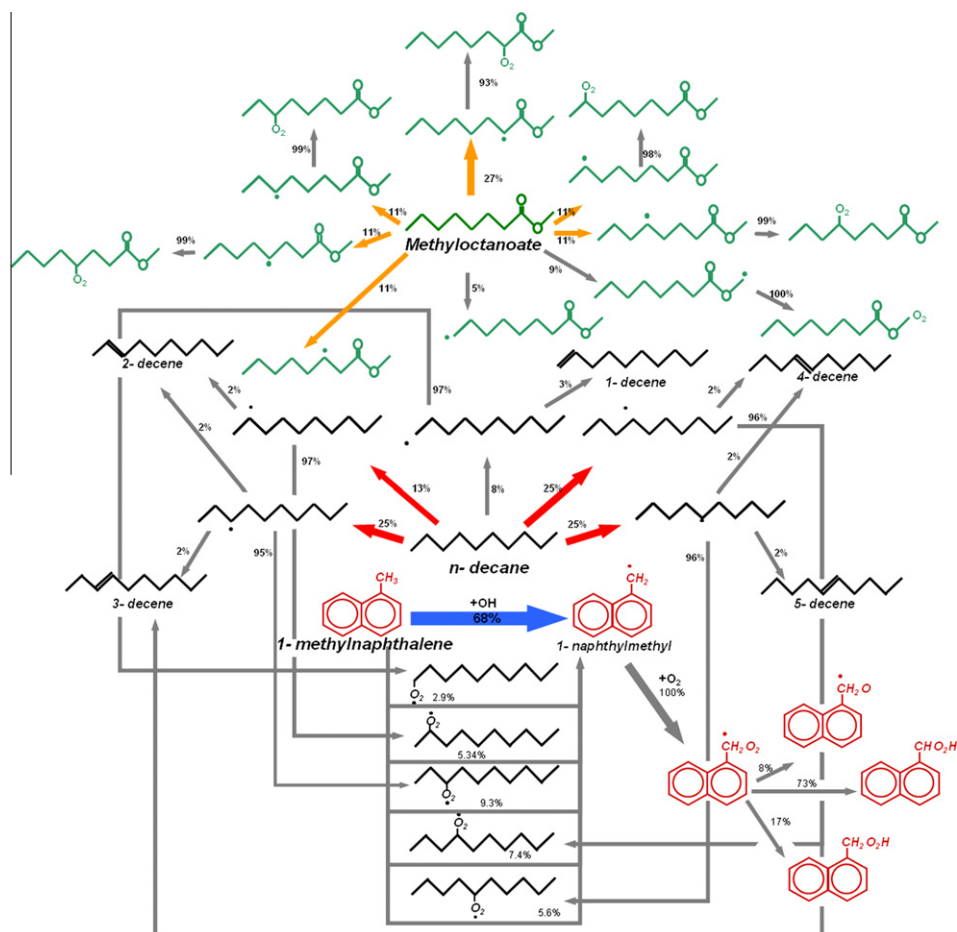


Fig. 9. Elemental reaction path analysis for the oxidation of B30 surrogate fuel at 640 K, 20 bar, 0.15% 1-methylnaphthalene, 0.011% *n*-Decane, and 0.01% methyloctanoate conversion under stoichiometric conditions. Main consumption reactions of primary fuel surrogate are represented by the color arrows: yellow for methyloctanoate, red for *n*-decane, and blue for 1-methylnaphthalene. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

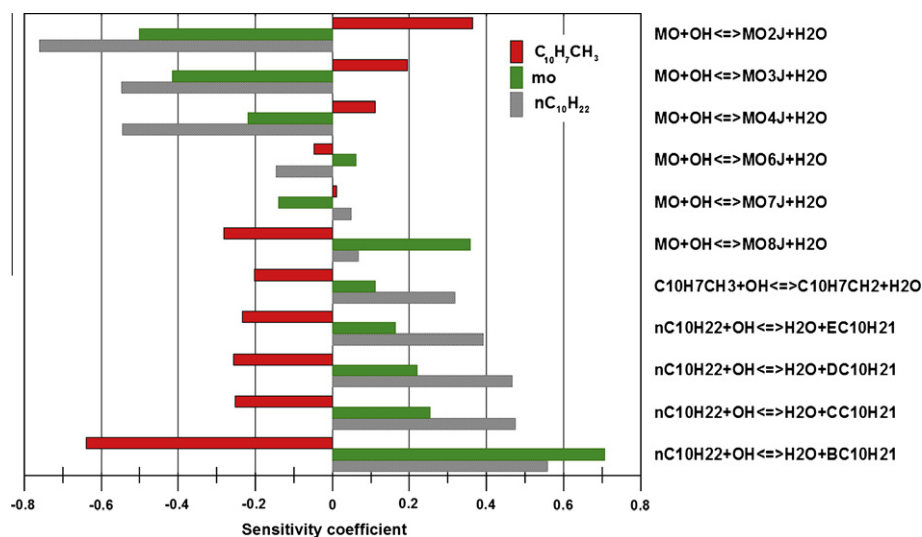


Fig. 10. Sensitivity analysis of the oxidation of B30 surrogate fuel at 640 K, $\phi = 1$, 20 bar, 0.15% 1-methylnaphthalene, 0.011% *n*-Decane, and 0.01% methyloctanoate conversion.

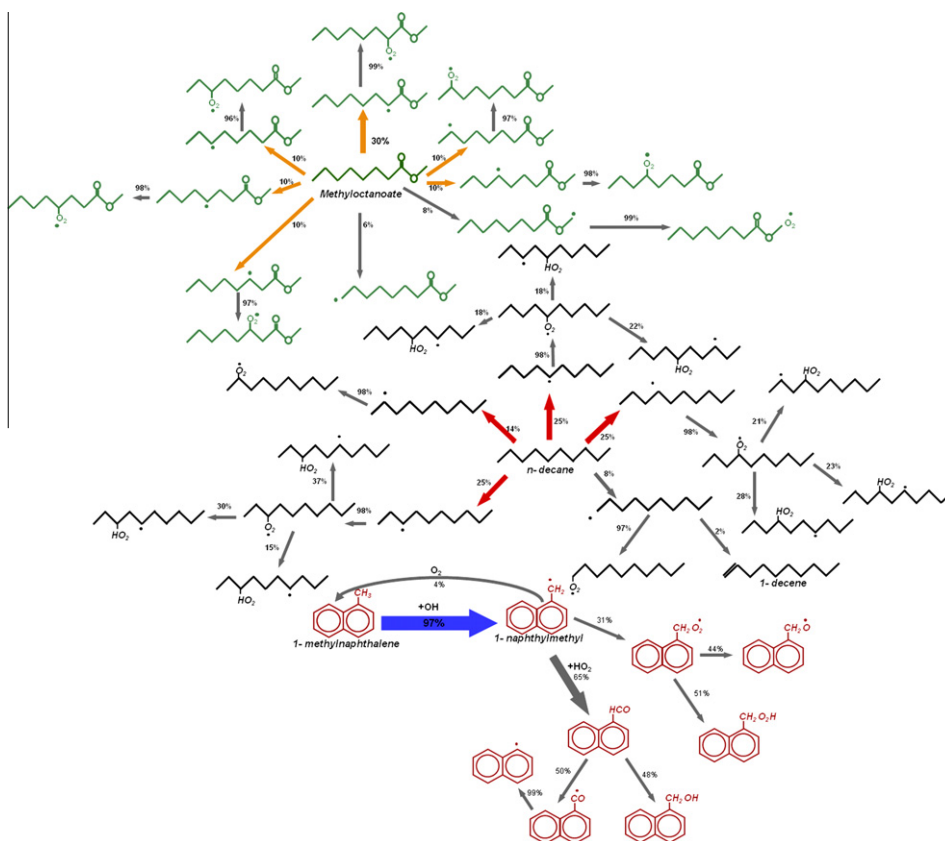


Fig. 11. Reaction path analysis for the oxidation of B30 surrogate fuel at 760 K, 20 bar, 39% 1-methylnaphthalene, 76% *n*-decane, and 13% methyl octanoate conversion under stoichiometric conditions. Main consumption reactions of primary fuel surrogate are represented by the color arrows: yellow for methyl octanoate, red for *n*-decane, and blue for 1-methylnaphthalene. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

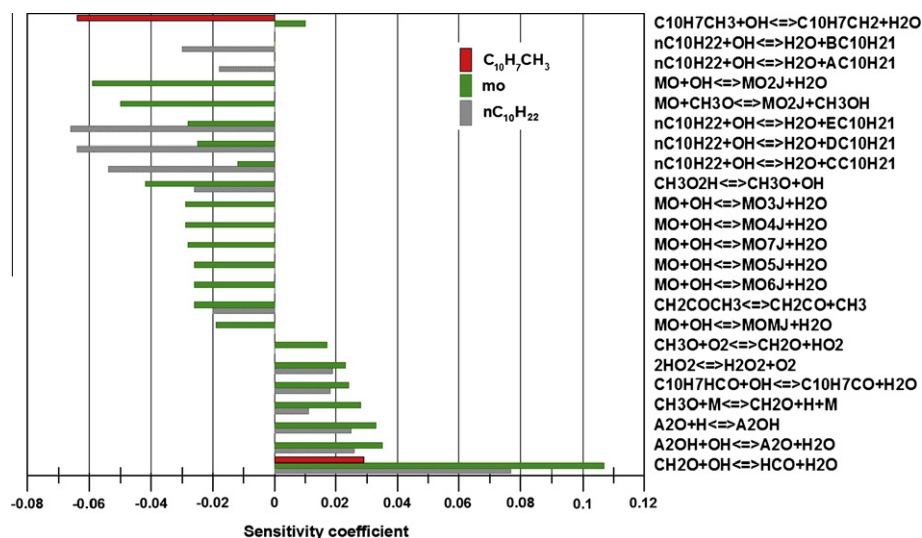


Fig. 12. Sensitivity analysis of the oxidation of B30 surrogate fuel at 760 K, $\phi = 1$, 20 bar, 39% 1-methylnaphthalene, 76% *n*-decane, and 13% methyl octanoate conversion.

pure *n*-decane, pure methyl octanoate, and pure 1-methylnaphthalene under similar JSR conditions.

4.2. Results of shock-tube experiments and modeling

In the present study, ignition delay times were evaluated from the CH^+ signals for the same B30 biodiesel fuel surrogate studied

before (i.e., $C_{9.91}H_{18.28}O_{0.6}$). The estimated uncertainties in the ignition delay time are in the range from ± 7.5 to $\pm 10\%$. Results and experimental conditions are listed in Table 3.

With all shock-tube data (Table 3), an Arrhenius correlation was developed for each equivalence ratio, for $T > 919$ K. Below that temperature, NTC behavior is observed and an Arrhenius fit could not be applied. Each fit establishes a relationship between the igni-

40 bar. As expected, fuel-lean mixtures are less reactive than fuel-rich mixtures. At the same equivalence ratio, high-pressure fuel/air mixtures ignite faster than at lower pressure.

As depicted in Fig. 7, adding the reactions given in Tables 1 and 2 to the mechanism improved the predictions of the ignition delay times. At low temperature ($\phi = 1.0$, 20 bar, and 880 K), the initial kinetic model overestimated the measurements with a maximum factor of 1.6. Conversely, under the same conditions, the new kinetic model improves the prediction to a maximum error of 30%.

Figure 8 also shows an excellent agreement between predictions and experiments at 20 bar and high temperature ($T > 970$ K) where $\tau(\text{exp.}) - \tau(\text{model})/\tau(\text{exp.})$ ranges from 0.15 to 0.23. However, at higher pressure, the model overestimates ignition delay times ($\tau(\text{exp.}) - \tau(\text{model})/\tau(\text{exp.})$ ranges from 1.06 to 1.64) across the same range of temperature ($T > 970$ K) and the effect of alkyl-oxygen bonds, found in the methyloctanoate structure, on ignition times cannot be simulated. Thus, even if simulations are in a good agreement with experimental data (Figs. 5 and 6), more validation is needed at low temperatures ($T < 900$ K) and high pressures ($p \geq 40$ bar) to improve the current mechanism.

It should be noted that the present model was also successfully tested against experimental ignition delay times available from the literature. For example, shock-tube data for pure *n*-decane from Pfahl et al. [22], at 13 bar and $\phi = 1$, and pure 1-methylnaphthalene from Wang et al. [21] at 10 bar and $\phi = 1$ could be well predicted by the proposed model (see the Supplementary material).

4.3. Reaction path and sensitivity analysis

In the following figures which present reaction path analysis results, the importance of every reaction is not given; we only present the reaction flux from a model-fuel component to each of the primary radicals. Figure 9 shows the most important oxidation pathways of the initial components of the B30 surrogate under stoichiometric conditions at 20 bar and 640 K. Although all experimental data were obtained at temperatures higher than 750 K, it is at a relatively lower temperature (when the autoignition process starts) where the chemical interactions between the different primary fuel components and their radicals take place. For that reason, the flux analysis at 640 K is interesting to perform. Figure 8 represents the major species and reaction pathways involved in the oxidation of 1-methylnaphthalene, *n*-decane, and methyloct-

anoate. It shows the complexity of the chemical processes taking place during hydrocarbon oxidation.

The main result from that reaction flux analysis is the evidenced importance of interactions of primary radicals with surrogate components or primary stable species, in particular those from *n*-decane and 1-methylnaphthalene (Table 2). The reaction scheme shows that primary radicals from first H-abstractions are mainly carried out by OH· radicals. At the same time, and with a smaller effect, the primary radicals are involved in H-abstraction reactions. Thus, as depicted in Fig. 9, these reactions consist of H-abstractions of 1-naphthylmethyl radicals from *n*-decane ($\text{C}_{10}\text{H}_{22} + \text{C}_{10}\text{H}_7\text{CH}_2 \rightleftharpoons \text{C}_{10}\text{H}_{21} + \text{C}_{10}\text{H}_7\text{CH}_3$) and of alkylperoxy radicals from 1-methylnaphthalene ($\text{C}_{10}\text{H}_7\text{CH}_3 + \text{C}_{10}\text{H}_{21}\text{O}_2 \rightleftharpoons \text{C}_{10}\text{H}_7\text{CH}_2 + \text{C}_{10}\text{H}_{21}\text{O}_2\text{H}$). Also, it can be seen from Fig. 9 that approximately 31% of 1-methylnaphthalene is consumed by H-abstractions initiated by the different decylperoxy radicals to form 1-naphthylmethyl. Around 98% of these alkylperoxy radicals come from *n*-decane. It can also be observed that most of the H-abstraction reactions involve OH· radicals. Further, 68% of the 1-methylnaphthalene is consumed to form the resonantly stabilized 1-naphthylmethyl radical, 96% of *n*-decane forms *n*-decyl, and 90% of methyloctanoate forms moXj ($\text{C}_7\text{H}_{14}\text{C}(\text{=O})\text{OCH}_3$) radicals. All of these primary radicals mainly form peroxy radicals by addition of O_2 : around 98% of moXOO·, 100% of $\text{C}_{10}\text{H}_7\text{CH}_2\text{OO} \cdot$, and around 95% of $\text{C}_{10}\text{H}_{21}\text{OO} \cdot$ are formed in this way. In the particular case of *n*-decane, 5% of decyl radicals produce olefins via oxidation by O_2 . These peroxy radicals can isomerize into the corresponding hydroperoxy radicals which in turn react via addition of a second O_2 molecule, as can be expected in this temperature range.

Reaction path analyses under these conditions have not shown unusually great changes in the modified mechanism compared to our initial mechanism. However, at this temperature, it was observed that the chemical pathways were slightly changed for *n*-decane: $\text{C}_{10}\text{H}_{22} + \text{C}_{10}\text{H}_7\text{CH}_2 \rightleftharpoons \text{C}_{10}\text{H}_{21} + \text{C}_{10}\text{H}_7\text{CH}_3$ (9.8% compared to 5.7% from previous mechanism) and 1-methylnaphthalene via $\text{C}_{10}\text{H}_7\text{CH}_3 + \text{OH} \rightleftharpoons \text{C}_{10}\text{H}_7\text{CH}_2 + \text{H}_2\text{O}$ (68.0% compared to 60.5%) and $\text{C}_{10}\text{H}_7\text{CH}_3 + \text{C}_{10}\text{H}_{21}\text{O}_2 \rightleftharpoons \text{C}_{10}\text{H}_7\text{CH}_2 + \text{C}_{10}\text{H}_{21}\text{O}_2\text{H}$ (30.5% compared to 27.2%). However, the importance of OH· radical reactions does not present major changes compared to the starting model.

Figure 10 depicts sensitivity coefficients for the B30 surrogate fuel primary components under the same conditions. According to the simulations, *n*-decane and methyloctanoate fractions are most sensitive to the recombination reactions of $n\text{C}_{10}\text{H}_{22}$ with

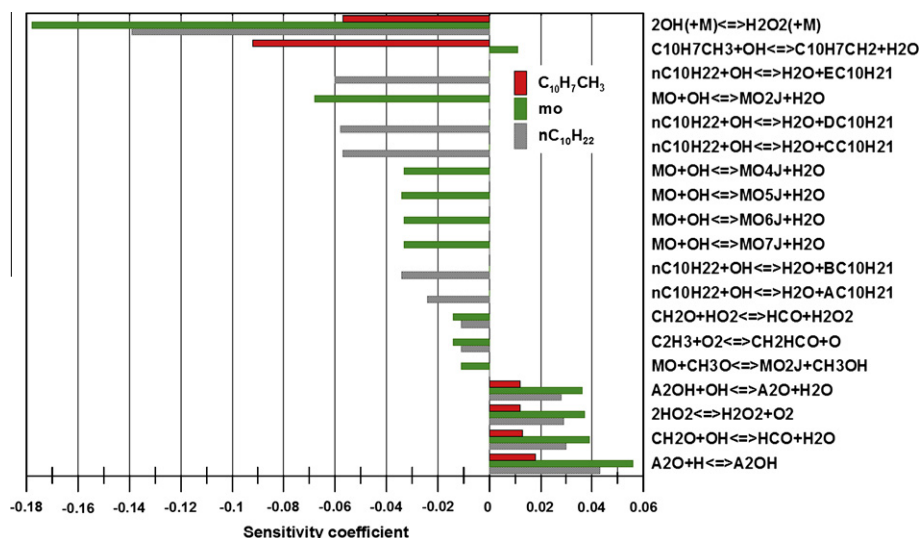


Fig. 14. Sensitivity analysis of the oxidation of B30 surrogate fuel at 910 K, $\phi = 1$, 20 bar, 22% 1-methylnaphthalene, 76% *n*-decane, and 64% methyloctanoate conversion.

$\text{OH}^\cdot$ and in an opposite way, these same reactions consume $\text{C}_{10}\text{H}_7\text{CH}_3$ via $n\text{C}_{10}\text{H}_{22} + \text{OH}^\cdot \rightleftharpoons \text{BC}_{10}\text{H}_{21} + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = 0.559$, $S_{\text{mo}} = 0.706$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = -0.339$), $n\text{C}_{10}\text{H}_{22} + \text{OH}^\cdot \rightleftharpoons \text{CC}_{10}\text{H}_{21} + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = 0.474$, $S_{\text{mo}} = 0.253$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = -0.253$), $n\text{C}_{10}\text{H}_{22} + \text{OH}^\cdot \rightleftharpoons \text{DC}_{10}\text{H}_{21} + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = 0.467$, $S_{\text{mo}} = 0.220$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = -0.256$), and $n\text{C}_{10}\text{H}_{22} + \text{OH}^\cdot \rightleftharpoons \text{EC}_{10}\text{H}_{21} + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = 0.391$, $S_{\text{mo}} = 0.163$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = -0.233$), as well as to the recombination of $\text{C}_{10}\text{H}_7\text{CH}_3$ and MO with $\text{OH}^\cdot$ via $\text{C}_{10}\text{H}_7\text{CH}_3 + \text{OH}^\cdot \rightleftharpoons \text{C}_{10}\text{H}_7\text{CH}_2 + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = 0.319$, $S_{\text{mo}} = 0.112$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = -0.202$) and $\text{MO} + \text{OH}^\cdot \rightleftharpoons \text{MO}_8 + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = 0.067$, $S_{\text{mo}} = 0.357$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = -0.282$), respectively. The overall reactivity of *n*-decane and methyloctanoate fractions is most sensitive to the recombination reactions of MO with $\text{OH}^\cdot$, and as before, in an opposite way, these same reactions are most sensitive to $\text{C}_{10}\text{H}_7\text{CH}_3$ via $\text{MO} + \text{OH}^\cdot \rightleftharpoons \text{MO}_2 + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = -0.760$, $S_{\text{mo}} = -0.502$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = 0.363$), $\text{MO} + \text{OH}^\cdot \rightleftharpoons \text{MO}_3 + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = -0.547$, $S_{\text{mo}} = -0.415$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = 0.195$) and $\text{MO} + \text{OH}^\cdot \rightleftharpoons \text{MO}_4 + \text{H}_2\text{O}$ ($S_{n\text{C}_{10}\text{H}_{22}} = -0.545$, $S_{\text{mo}} = -0.219$, and $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = 0.111$). An increase of the rate constants of this reaction would accelerate fuel consumption and decrease the ignition delay time. It can be noted that at 640 K only two primary components of the B30 surrogate fuel (*n*-decane and methyloctanoate) show the same sensitive behavior, contrary to 1-methylnaphthalene, which is oxidized thanks to the radical pool provided through the oxidation of *n*-decane and methyloctanoate at low temperature.

During the present study, we could also observe that these cross-reactions become less important when temperature increases. This is due to a decrease in the concentration of the largest radicals as temperature increases. Therefore, reactions between smaller radicals become more important as we can see from our

investigations at 760, 910, and 1210 K at 20 bar and under stoichiometric conditions (Figs. 11, 13 and 15).

At the beginning of the negative temperature coefficient (NTC) regime (conditions shown in Fig. 11), the fuel undergoes H-abstraction mainly by $\text{OH}^\cdot$ and $\text{O}^\cdot$ radicals to produce primary radicals. For example, the consumption route of *n*-decane consists of ca. 92% H-abstraction by $\text{OH}^\cdot$ to produce decyl radicals. At the same time, 1-methylnaphthalene produces mainly naphthylmethyl (97% via reactions with $\text{OH}^\cdot$), which is followed by alkylperoxy formation and then by the summarization of alkylhydroperoxy radicals. The most important routes of consumption of methyloctanoate are H-abstraction by $\text{OH}^\cdot$ (ca. 88%) to give moXj radicals. At this temperature, most primary radicals react by addition of O_2 . Thus, around 97% of decyl and moXj radicals form alkylperoxy by addition of O_2 . Additionally, 1-naphthylmethyl presented different behavior because only 31% reacts with O_2 to produce $\text{C}_{10}\text{H}_7\text{CH}_2\text{OO}^\cdot$ and 65% reacts with HO_2 to produce 1-naphthylaldehyde. Finally, alkylperoxy radicals isomerize into alkylhydroperoxy radicals before starting the branching process ($\text{RO}_2^\cdot \rightleftharpoons \text{QOOH}$ and $\text{OOQO}_2\text{H}^\cdot \rightarrow \text{HOOQO}_2\text{H}$).

Concerning fuel consumption, only the oxidation routes of 1-methylnaphthalene are modified by the updated kinetics. The consumption of 1-methylnaphthalene via $\text{C}_{10}\text{H}_7\text{CH}_3 + \text{OH}^\cdot \rightleftharpoons \text{C}_{10}\text{H}_7\text{CH}_2 + \text{H}_2\text{O}$ increases from 75% (starting mechanism) to 94%.

Sensitivity analysis results for the oxidation of B30 surrogate fuel under these conditions are presented in Fig. 12. In this case, 1-methylnaphthalene becomes sensitive to the reaction involving H-abstraction by $\text{OH}^\cdot$ ($\text{CH}_2\text{O}^\cdot + \text{OH}^\cdot \rightleftharpoons \text{HCO}^\cdot + \text{H}_2\text{O}$; $S_{\text{C}_{10}\text{H}_7\text{CH}_3} = 0.029$). This reaction also reduces the reactivity for *n*-decane and methyloctanoate at the beginning of the NTC region ($S_{n\text{C}_{10}\text{H}_{22}} = 0.077$, and

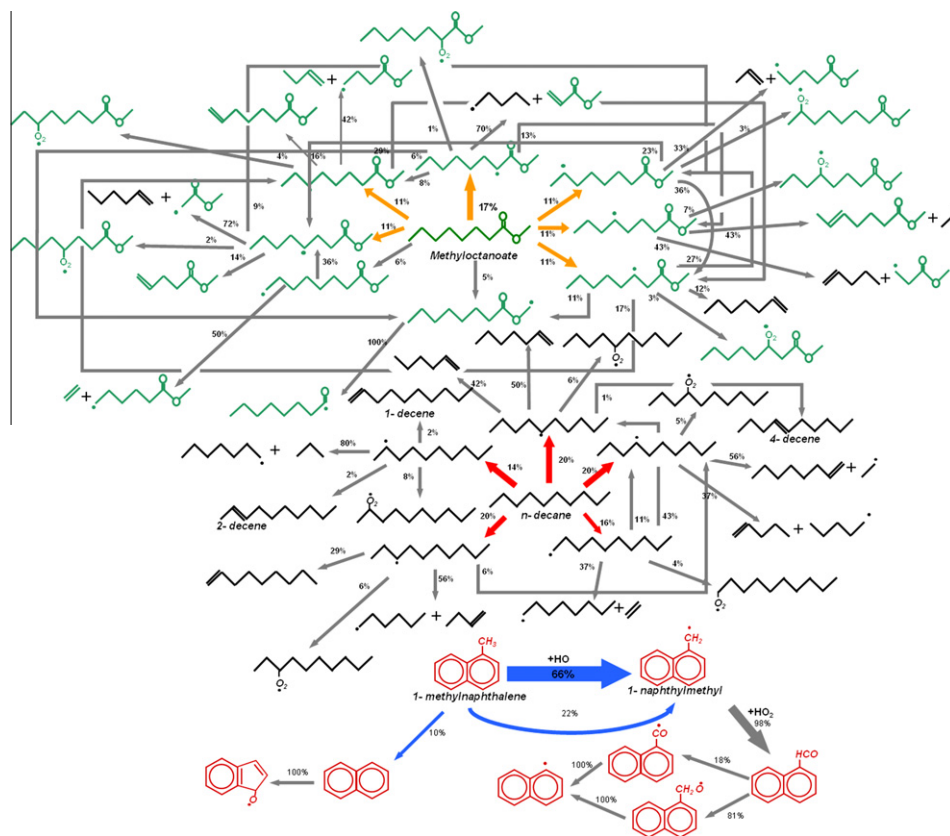


Fig. 15. Reaction path analysis for the oxidation of B30 surrogate fuel at 1210 K, 20 bar, 19% 1-methylnaphthalene, 46% *n*-decane, and 61% methyloctanoate conversion under stoichiometric conditions. Main consumption reactions of primary fuel surrogate are represented by the color arrows: yellow for methyloctanoate, red for *n*-decane, and blue for 1-methylnaphthalene. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$S_{mo} = 0.107$) because it consumes $OH^\cdot$ radicals at the expense of the fuel components. According to these results, 1-methylnaphthalene, *n*-decane, and methyloctanoate are mainly consumed via H-abstraction by $OH^\cdot$ ($S_{C_{10}H_7CH_3} = -0.064$, $S_{nC_{10}H_{22}} = -0.066$, and $S_{mo} = -0.059$), giving the first primary radicals.

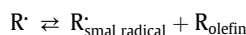
Similar behavior, with different flux intensities, was observed at the end of the NTC region (Fig. 13). As expected, the main fuel consumption paths at this temperature are H-atom abstraction by small radicals. Specifically, at 910 K and under the same conditions (20 bar and $\phi = 1$), the main routes of consumption of *n*-decane and 1-Methylnaphthalene are H-abstraction by $OH^\cdot$ and $O^\cdot$ radicals to produce decyl and 1-naphthylmethyl radicals. 1-Methylnaphthalene also produces naphthalene (1.3% via $H^\cdot$). Then naphthylmethyl produces 1-naphthaldehyde (71% via reaction with $HO_2^\cdot$). These reactions are followed by peroxide formation and then by the summarization of hydroperoxy. At the same time, the main routes of consumption of methyloctanoate are H-abstraction by $H^\cdot$ (1.2%), $CH_3O^\cdot$ (1.3%), and $OH^\cdot$ (81.3%) to give moXj radicals. Therefore, most of the primary radicals react by addition of O_2 : between 81% and 97% of moXj radicals yield moXoo and around 96% of decyl yield decylperoxy through addition of O_2 . 1-Naphthylmethyl has the same behavior as before: only 25% reacts with O_2 to produce $C_{10}H_7CH_2OO^\cdot$ and 71% reacts with $HO_2^\cdot$ to produce 1-naphthaldehyde.

Concerning $OH^\cdot$ production at 910 K, as before, only a 1% increase was observed with the modified mechanism compared to the initial mechanism. The main reactions responsible of $OH^\cdot$ production with the NTC regime are: $2OH^\cdot (+M) \rightleftharpoons H_2O_2 (+M)$, $CH_3 + HO_2^\cdot \rightleftharpoons CH_3O^\cdot + OH^\cdot$, and $QOOH \rightleftharpoons QO^\cdot + OH^\cdot$.

Figure 14 depicts sensitivity coefficients of fuel components under the same conditions at the high-temperature end of the NTC region. Here, we can also observe H-abstraction by $OH^\cdot$ from fuel components yielding primary radicals. However, fuel consumption is most sensitive to the reaction involving decomposition of hydrogen peroxide to generate two $OH^\cdot$ radicals ($S_{C_{10}H_7CH_3} = -0.139$, $S_{nC_{10}H_{22}} = -0.057$, and $S_{mo} = -0.178$). Conversely, the formation reaction of A_2OH via $A_2O^\cdot + H^\cdot \rightleftharpoons A_2OH$ is the most inhibiting reaction ($S_{C_{10}H_7CH_3} = 0.043$, $S_{nC_{10}H_{22}} = 0.018$, and $S_{mo} = 0.056$).

It is also interesting to observe the kinetic behavior at high temperature. To that end, a reaction pathway analysis was performed at 1210 K and 20 bar under stoichiometric conditions (Fig. 14).

Under these conditions, H-abstraction mainly occurs with $H^\cdot$, $CH_3^\cdot$, $O_2^\cdot$, and $O^\cdot$. These reactions become more significant than those observed in the NTC cases (760 and 910 K) presented before. We can also see an increase in radical summarization resulting in a decreased formation of alkyl peroxides. The dominant reactions are cracking reactions yielding light hydrocarbons. H-atom abstraction from *n*-decane occurs by reaction with $OH^\cdot$ (59.3%), $H^\cdot$ (17.6%), $O^\cdot$ (8.8%), and $CH_3^\cdot$ (3.1%) producing decyl radicals. H-atom abstractions from methyloctanoate mainly occur by reaction with $H^\cdot$ (23.8%) and $OH^\cdot$ (43.8%), giving moXj radicals. 1-Naphthylmethyl is formed by H abstraction of 1-methylnaphthalene by $OH^\cdot$ (66%), $H^\cdot$ (14%), $O_2^\cdot$ (5%), and $CH_3^\cdot$ (3%). Naphthalene is formed by reactions of $H^\cdot$ (10%) with 1-methylnaphthalene. *n*-decane primary radicals undergo C–C bond breaking (β -scission), producing large olefins ($\approx 30\%$) and small olefins plus small radicals ($\approx 50\%$). Also, by O_2 addition, decyl radicals convert into alkyl peroxides to a lesser extent than in the case studied before ($\approx 6\%$). Radical summarization (decyl and moXj) by H transfer is also observed. Along with methyloctanoate C–C bond breaking by β -scission, small olefins, small acid radicals (ca. 30–50%), and moXoo ($\approx 5\%$) are produced. Different behavior was observed for 1-naphthylmethyl radicals, for which the O_2 addition reaction becomes less important, while the main reaction proceeds with $HO_2^\cdot$ (98%) to produce 1-naphthaldehyde:



Consumption of methyloctanoate via $mo + OH^\cdot \rightleftharpoons moXj + H_2O$ was affected by increased OH production (by $\approx 1\%$ compared to the initial mechanism). The same behavior was observed for 1-methylnaphthalene, consumption of which increased by $\approx 1\%$ via $C_{10}H_7CH_3 + OH^\cdot \rightleftharpoons C_{10}H_7CH_2^\cdot + H_2O$, while that of *n*-decane via $nC_{10}H_{22} \rightleftharpoons OH^\cdot + BC_{10}H_{21}$ increased by $\approx 3\%$. However, $OH^\cdot$ production occurs via the same routes as before. The main reactions yielding OH at 1210 K are $2OH^\cdot (+M) \rightleftharpoons H_2O_2 (+M)$, $H^\cdot + HO_2^\cdot \rightleftharpoons 2OH^\cdot$, and $CH_3 + HO_2^\cdot \rightleftharpoons CH_3O^\cdot + OH^\cdot$.

Figure 16 presents sensitivity analysis results for the oxidation of the B30 surrogate. As expected, sensitivity analyses at 1210 K have not revealed unusual behavior of high-temperature chemistry. The system is sensitive to H-abstraction by $OH^\cdot$ from fuel components and to the kinetics of the C_0 – C_1 submechanism, i.e.,

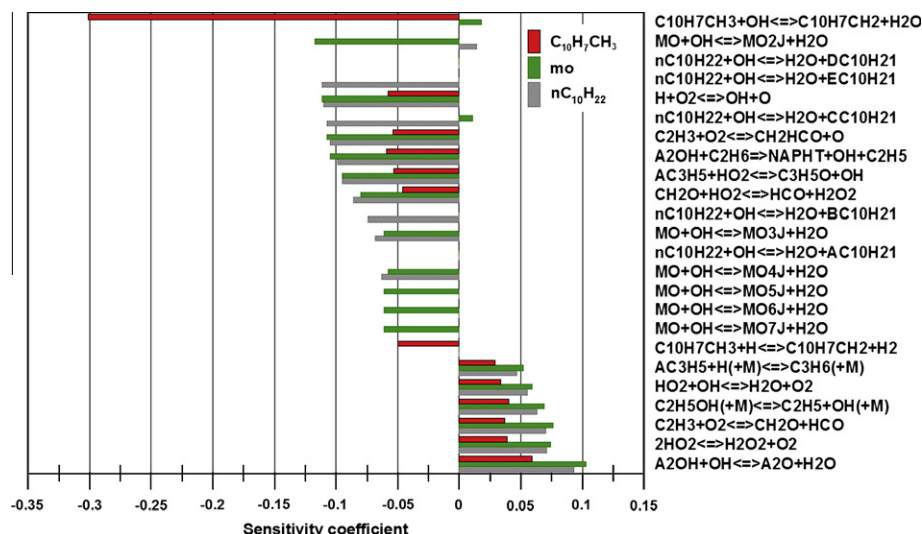
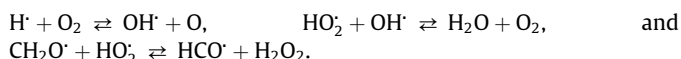


Fig. 16. Sensitivity analysis of the oxidation of B30 surrogate fuel at 1210 K, $\phi = 1$, 20 bar, 19% 1-methylnaphthalene, 46% *n*-decane, and 61% methyloctanoate conversion.



5. Conclusions

Ignition delay times of a three-component B30 surrogate fuel consisting of 49% *n*-decane, 30% methyl octanoate, and 21% 1-methylnaphthalene in mol% were measured in a shock tube behind a reflected wave at pressures of 20 and 40 bar, of equivalence ratios of 0.5–1.5, and at post-reflected-shock temperatures of $700 < T_5 < 1200$ K.

A detailed chemical kinetic reaction mechanism for biodiesel B30 autoignition was developed and validated. It was also tested against experimental data from a jet-stirred reactor [2]. The new model can describe shock-tube ignition delay times for a B30 surrogate fuel reasonably well over a broad range of conditions (700–1200 K, 20 and 40 bar, equivalence ratios of 0.5–1.5). In addition, it was found that the proposed mechanism predicts very well experimental data obtained by Pfahl et al. [22] and Wang et al. [21].

The main reaction paths were delineated, showing the effect of new reactions on the production and consumption of OH radicals. The modeling also showed that the interactions between the fuel components were more significant at lower temperature (i.e., $T = 640$ K). These findings need further confirmation. Finally, model validation against flame speed data would be useful. This work is under way.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2011.10.006.

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